

Challenge: Detect Outliers with IQR

Detecting outliers is a crucial part of data cleaning, as these extreme values can distort your analysis and lead to misleading conclusions. One common and robust approach for identifying outliers in a numerical column is the **interquartile range (IQR) method**. This method is especially useful because it is not affected by extreme values, unlike methods based on the mean and standard deviation.

The IQR method works by first calculating the **first quartile (Q1)** and the **third quartile (Q3)** of the data. Q1 is the value below which 25% of the data fall, and Q3 is the value below which 75% of the data fall. The **IQR itself is simply Q3 minus Q1**, representing the range of the middle 50% of your data.

Once you have the IQR, you can define the lower and upper bounds for typical values in your dataset. Any data point that falls below Q1 minus 1.5 times the IQR, or above Q3 plus 1.5 times the IQR, is considered an **outlier**. This rule is widely used because it provides a balance between sensitivity and robustness.

```
import pandas as pd

# Create a DataFrame with a numerical column containing some outliers
data = {
    "score": [10, 12, 13, 14, 15, 16, 17, 18, 100, 110]
}
df = pd.DataFrame(data)

print("Original DataFrame:")
print(df)
```